

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250273524

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

WINKLER; Anton et al.

INSULATION FOR MAGNETIC-MOLDING COMPOUND (MMC) MODULE

Abstract

A module includes: an integrated circuit (IC) die; a leadframe having contact pads; solder elements coupled between the IC die and the contact pads; an insulative layer between the solder elements; and magnetic-molding compound (MMC) filler between the IC die and the leadframe.

Inventors: WINKLER; Anton (San Jose, CA), INOUE; Hidetoshi (Oyama-Shi, JP), OTAKE; Kenji (Karuizawa, JP)

Applicant: TEXAS INSTRUMENTS INCORPORATED (Dallas, TX)

Family ID: 1000006764265

Appl. No.: 18/050458

Filed: October 27, 2022

Publication Classification

Int. Cl.: H01L23/29 (20060101); H01L21/48 (20060101); H01L21/56 (20060101); H01L23/31 (20060101); H01L23/495 (20060101)

U.S. Cl.:

CPC H01L23/295 (20130101); H01L21/4825 (20130101); H01L21/565 (20130101); H01L23/3107 (20130101); H01L23/49503 (20130101); H01L23/4952 (20130101); H10D1/20 (20250101);

Background/Summary

BACKGROUND

[0001] Many integrated circuits (IC) are designed for use with external components. For example, an IC may include contact pads to couple IC circuitry to external ICs or passive components (e.g., resistors, capacitors, and/or inductors). To enhance energy storage of an inductor, a magnetic core may be used. In a conventional approach, a module having an IC die and an external inductor uses magnetic-molding compound (MMC) filler to form the magnetic core of the external inductor. However, some of the metallic particles of the MMC filler may result in undesirable leakage current between the IC die and other components or metallic contacts of the module.

SUMMARY

[0002] In an example embodiment, a module comprises: an integrated circuit (IC) die; a leadframe having contact pads; solder elements coupled between the IC die and the contact pads; an insulative layer between the solder elements; and magnetic-molding compound (MMC) filler between the IC die and the leadframe.

[0003] In another example embodiment, a switching converter device comprises: an IC die including switching converter components; a leadframe having contact pads; solder elements coupled between the IC die and the contact pads; and an insulative layer between the solder elements. The switching converter device also comprises a coil having a core and first and second ends. The first end is coupled to a first solder element of the solder elements and a respective contact pad of the contact pads. The second end is coupled to a second solder element of the solder elements and a respective contact pad of the contact pads. The switching converter device also comprises magnetic-molding compound (MMC) filler in the core and between the IC die and the leadframe.

[0004] In yet another example embodiment, a method of manufacturing a module comprises: obtaining an IC die; obtaining a leadframe having contact pads; obtaining a coil having a core and first and second ends; adding an insulative layer to partially cover the contact pads; adding solder elements to exposed portions of the contact pads; and adding magnetic-molding compound (MMC) filler in the core and between the IC die and the leadframe.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a perspective view showing a module in accordance with an example embodiment.

[0006] FIGS. 2A-2H are cross-sectional views showing assembly of the module of FIG. 1 in accordance with an example embodiment.

[0007] FIGS. 3A and 3B are top view of insulation layer layout options in accordance with different example embodiments.

[0008] FIG. 4 is cross-sectional view showing layers of the module of FIG. 1 in accordance with an example embodiment.

[0009] FIG. 5A is a diagram showing MMC resistivity for a module without an insulative layer in accordance with a conventional approach.

[0010] FIG. 5B is a diagram showing MMC resistivity for a module with an insulative layer in accordance with an example embodiment.

DETAILED DESCRIPTION

[0011] The same reference numbers or other reference designators are used in the drawings to designate the same or similar (either by function and/or structure) features.

[0012] FIG. 1 is a perspective view showing a module **100** in accordance with an example embodiment. In the example of FIG. 1, the module **100** includes a leadframe **102** having conductive channels **104** and contact pads, including contact pads **106A** and **106B** and a set of contact pads **106C**. In some example embodiments, the leadframe **102** is a routable leadframe

(RLF), in which the conductive channels **104** relative to the contact pads **106A** and **106B** and the set of contact pads **106C** are routable during fabrication of the leadframe **102**. In the example of FIG. **1**, the contact pad **106A** couples to a first end **110** of a coil **108**. The contact pad **106B** coupled to a second end **112** of the coil **108**. The set of contact pads **106C** couple to respective contacts (not shown) of an IC die **116** included with the module **100**.

[0013] In the example of FIG. **1**, the module **100** also includes solder elements **114A** and **114B** and a set of solder elements **114C**. Specifically, the solder element **114A** is coupled between the contact pad **106A** and the first end **110** of the coil **108**. The solder element **114B** is coupled between the contact pad **106B** and the second end **112** of the coil **108**. The set of solder elements **114C** are coupled between the set of contact pads **106C** and respective contacts of the IC die **116**. As needed, the solder elements **114A** and **114B**, and the set of solder elements **114C** may be supplemented using conductive leads (e.g., conductive leads **214** in FIGS.) to extend the conductive connection between the contact pads **106A** and **106B** and the coil **108** and/or to extend the conductive connection between the set of contact pads **106C** and the IC die **116**.

[0014] In the example of FIG. **1**, there is a gap **113** between the leadframe **102** and the IC die **116**, where the gap **113** is filled with a magnetic-molding compound (MMC) filler **120**. The MMC filler **120** also fills a core of the coil **108**. Without limitation, the MMC filler **120** forms a three-dimensional housing (e.g., a rectangular or cubic prism) around all other components of module **100** except the leadframe **102**, which forms the base of the module **100**. In some example embodiments, the thickness of the gap **113** is selected based on a target gap that limits MMC filler particle size between the IC die **116** and the leadframe **102** to less than a threshold particle size. In some example embodiments, the target gap between the IC die **116** and the leadframe **102** is achieved by adding an insulative layer **122** to the leadframe **102** in the area of the set of contact pads **106C**. Additionally or alternatively, an insulative layer could be added to IC die **116** so that the gap **113** is less than the target gap. In either case, the thickness of the insulative layer **122** limits the thickness of the MMC filler **120** between the IC die **116** and the leadframe **102**.

[0015] Without limitation, an example threshold particle size is 25 μm . In such case, the target gap between the IC die **116** and the leadframe **102** will be 25 μm . In this example, if the gap **113** between the set of contact pads **106C** of the leadframe **102** and IC die **116** is 40 μm , a 15 μm thickness for the insulation layer **122** achieves the target gap. With the module **100**, the MMC filler **120** improves the energy storage capacity of the coil **108**. By insulating the set of contact pads **106C** using the controlling the gap **113** and thus the thickness of the MMC filler **120** between the leadframe **102** and IC die **116**, leakage current is reduced compared to a conventional approach.

[0016] In an example scenario, MMC filler particles may increase leakage current between ground (GND) of the IC die **116** and input voltage (VIN) contact pads (e.g., the set of contact pads **106C**) of the leadframe **102**. Leakage current over time may cause defects such as early failure of a module (e.g., the module **100**). In some example embodiments, leakage current is suppressed by increasing insulation of the IC die **116** and/or contact pads (e.g., the set of contact pads **106C**) of the leadframe **102**. Gap reduction (between the IC die **116** and the leadframe **102**) or other MMC filter particle size control options may also help suppress leakage current.

[0017] FIGS. **2A-2H** are cross-sectional views **202**, **204**, **206**, **208**, **210**, **212**, **216**, and **218** showing assembly of the module **100** of FIG. **1** in accordance with an example embodiment. In the cross-sectional view **202** of FIG. **2A**, the leadframe **102** is prepared. In some example embodiments, preparing of the leadframe **102** includes adding and/or routing conductive channels such as the conductive channels **104** in FIG. **1**. As shown, the leadframe **102** includes contact pads **106A** and **106B**, and the set of contact pads **106C**. During preparation of the leadframe **102**, the position of the contact pads **106A** and **106B**, and the set of contact pads **106C** is selected and/or the connection of the contact pads **106A** and **106B**, and the set of contact pads **106C** to respective conductive channels is performed.

[0018] In the cross-sectional view **204** of FIG. **2B**, the insulative layer **122** is applied to the

leadframe **102**. In some example embodiments, the insulative layer **122** is applied so that it initially covers the set of contact pads **106C** without covering the contact pads **106A** and **106B**. As another option, the insulative layer **122** may initially cover the contact pads **106A** and **106B** as well as the set of contact pads **106C**. The thickness of the insulative layer **122** is selected to achieve a target insulation and/or a target gap between the leadframe **102** (with the insulative layer **122**) and the IC die **116**.

[0019] In the cross-sectional view **206** of FIG. 2C, the insulative layer **122** is modified, resulting in a modified insulative layer **122***. As shown, the modified insulative layer **122*** at least partially exposes the set of contact pads **106C**, while insulating remaining portions of the set of contact pads **106C**. In some example embodiments, modified insulative layer **122*** is based on application of an etching process or another targeted material removal process to the insulative layer **122**.

[0020] In the cross-sectional view **208** of FIG. 2D, the solder elements **114A** and **114B** and a set of solder elements **114C** are applied. Specifically, the solder element **114A** is applied to the exposed portion of the contact pad **106A**. The solder element **114B** is applied to the exposed portion of the contact pad **106B**. The set solder elements **114C** are applied to the exposed portions of the set contact pads **106C**.

[0021] In the cross-sectional view **210** of FIG. 2E, the IC die **116** and the coil **108** are positioned. Also, a set of conductive leads **214** (e.g., conductive posts) are positioned between set of solder elements **114C** and the IC die **116**. After positioning the coil **108**, the first end **110** of the coil **108** is aligned with the solder element **114A** and contact pad **106A**, while the second end **112** of the coil **108** is aligned with the solder element **114B** and contact pad **106B**. Also, the set of solder elements **114C** as well as contacts of the IC die **116** will be aligned with the conductive leads **214**. The cross-sectional view **210** also shows reduction of gap thickness from an initial gap **211** to the gap **113** due to the modified insulative layer **122*** being used.

[0022] In the cross-sectional view **212** of FIG. 2F, solder reflow is performed by heating the module components. The solder reflow process results in: the contact pad **106A** being coupled to first end **110** of the coil **108** by the solder element **114A**; the contact pad **106B** being coupled to second end **112** of the coil **108** by the solder element **114B**; the set of contact pad **106C** being coupled to the IC die **116** by the set of solder element **114C** and the set of conductive leads **214**; and/or other solder connections of the module being completed.

[0023] In the cross-sectional view **216** of FIG. 2G, the MMC filler **120** is applied. The MMC filler **120** is applied, for example, by using a mold (not shown) to form a three-dimensional housing (e.g., a rectangular or cubic prism) around all other components of module **100** except the leadframe **102**, which forms the base of the module **100**. For example, the MMC filler **120** may initially flow or be in a liquid state before hardening into a predetermined shape based on the mold used. After MMC filler application is complete, the IC die **116** is separated from the leadframe **102** by the gap **113** whose thickness (and related thickness of the MMC filler **120** in the gap **113**) is determined in part by a thickness of the modified insulation layer **122***.

[0024] In the cross-sectional view **218** of FIG. 2H, dicing operations are performed by a dicer **220** to separate the module **100** from a printed circuit board (PCB) or other leadframe assembly. In some example embodiments, many such modules are assembled together before the dicing operations are performed. After the dicing operations, the module **100** is complete. As desired, the module **100** is subsequently provided to customers or otherwise coupled to related circuitry to provide a desired function (e.g., switching converter operations).

[0025] FIGS. 3A and 3B are top views **300** and **310** of insulation layer layout options in accordance with different example embodiments. In the top views **300** and **310**, the contact pads **106A** and **106B**, and the set of contact pads **106C** are represented. As shown, the layouts for a first insulation layer **122A** and a second insulation layer **122B** may vary. More specifically, the first insulation layer **122A** in FIG. 3A has a rectangle layout (e.g., a geometric shape) that includes the set of contact pads **106C**, while the second insulation layer **122B** in FIG. 3B has an amorphous layout

(e.g., a non-geometric shape) that includes the set of contact pads **106C**. The layout option of the first insulation layer **122A** may be selected, for example, to simplify application of an insulation layer to a leadframe to reduce costs. The layout option of the second insulation layer **122B** may be selected, for example, to customize application of an insulation layer to a MMC module scenario (e.g., to reduce delamination between the insulation layer and MMC filler).

[0026] FIG. **4** is cross-sectional view showing layers of the module **100** of FIG. **1** in accordance with an example embodiment. As shown, the layers of the module **100** include layers of the leadframe **102** such as: some of the conductive channels **104**; some of the contact pads of the set of contact pads **106C**; conductive material **406** between conductive channels **104** and respective contact pads of the set of contact pads **106C**. Without limitation, the conductive channels **104**, the conductive material **406**, and the set of contact pads **106C** are made from copper (Cu) or a copper alloy. In some example embodiments, the remaining layers or portion of the leadframe **102** include Ajinomoto Build-Up Film@ (ABF) **402**.

[0027] In the example of FIG. **4**, the leadframe **102** includes the modified insulative layer **122*** over partially exposed portions of the set of contact pads **106C**. In some example embodiments, the modified insulative layer **122*** is a solder resist layer. The set of solder elements **114C** couple to the exposed portions of the set of contact pads **106C**. Other layers of the module **100** of FIG. **4** include the IC die **116**, which may include circuitry layers (not shown) and a polyimide (PI) layer **404** for additional insulation. Between the IC die **116** and the solder elements **114C** are the conductive leads **214**. In some example embodiments, the conductive leads **214** are copper posts.

[0028] In the example of FIG. **4**, MMC filler is represented using a first set of MMC filler particles **120A** and a second set of MMC filler particles **120B**. As shown, the MMC filler particles of the second set of MMC filler particles **120B** are smaller than the MMC filler particles of the first set of MMC filler particles **120A**. Due to the modified insulative layer **122***, the gap **113** between the leadframe **102** and the IC die **116** is limited such that only the second set of MMC filler particles **120B** fill the gap **113** (the first set of MMC filler particles **120A** are too large to fit in the gap). By adjusting a thickness of the modified insulative layer **122***, the gap **113** is adjustable. The modified insulative layer **122*** also limits exposure of the contact pads **106C**. With the modified insulative layer **122***, the PI layer **404**, and/or the second set of MMC filler particles **120B** in the gap **113** between the leadframe **102** and the IC die **116**, leakage current is reduced compared to a conventional approach.

[0029] FIG. **5A** is a diagram **500** showing a leakage current path **502** for a module without an insulative layer (e.g., the insulative layer **122** of FIGS. **1** and **2**, or the modified insulative layer **122*** of FIGS. **2** and **4**) in accordance with a conventional approach. As shown in the diagram **500** of FIG. **5A**, the leakage current path **502** flows between a first conductive lead **214A** and a second conductive lead **214B** using some of the first set of MMC filler particles **120A**, resulting in a shorter leakage current path (compared to the leakage current path **512** in FIG. **5B**). This is due, at least in part, to the initial gap **211** being larger than the size of the first set of MMC filler particles **120A**. In the example of FIG. **5A**, the first conductive lead **214A** carries an input voltage (VIN) and the second conductive lead **214B** carries an output voltage (VOUT) (e.g., related to a switching converter).

[0030] FIG. **5B** is a diagram **510** showing a leakage current path **512** for a module (e.g., the module **100** of FIGS. **1**, **2**, and **4**) having an insulative layer in accordance with an example embodiment. As shown in the diagram **510** of FIG. **5B**, the leakage current path **512** flows between the first conductive lead **214A** and the second conductive lead **214B** using some of the second set of MMC filler particles **120B**, resulting in a longer leakage current path (compared to the leakage current path **502** in FIG. **5A**). In the example of FIG. **5B**, the addition of the modified insulative layer **122*** and the PI layer **404** provides insulation and limits the size of MMC filler particles in the gap **113** between to the second set of MMC filler particles **120B** (the second set of MMC filler particles **120A** are avoided in the gap **113**). By adding insulative layers and/or limiting MMC particle size as

in the example of FIG. 5B, leakage current is reduced

[0031] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0032] Also, in this description, the recitation “based on” means “based at least in part on.”

Therefore, if X is based on Y, then X may be a function of Y and any number of other factors.

[0033] A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

[0034] As used herein, the terms “terminal”, “node”, “interconnection”, “pin” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component.

[0035] A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

[0036] Circuits described herein are reconfigurable to include additional or different components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the resistor shown. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor.

[0037] While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other example embodiments, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated. As used herein, the term “integrated circuit” means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.

[0038] In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within ± 10 percent of that parameter.

[0039] Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. A module, comprising: an integrated circuit (IC) die; a leadframe having contact pads; solder elements coupled between the IC die and the contact pads; an insulative layer between the solder elements; and magnetic-molding compound (MMC) filler between the IC die and the leadframe.
2. The module of claim 1, wherein a thickness of the MMC filler between the IC die and the leadframe is limited by a thickness of the insulative layer.
3. The module of claim 2, wherein the thickness of the insulative layer is based on a target gap between the IC die and the leadframe.
4. The module of claim 3, wherein the target gap is selected to limit MMC filler particle size between the IC die and the leadframe to less than a threshold particle size.
5. The module of claim 1, wherein the insulative layer comprises solder resist.
6. The module of claim 1, wherein the insulative layer partially exposes the contact pads.
7. The module of claim 1, wherein the insulative layer has a layout based on an outline of the contact pads.
8. The module of claim 1, further comprising a coil having a first end and a second end, the first end coupled to a first solder element of the solder elements and a respective contact pad of the contact pads, the second end coupled to a second solder element of the solder elements and a respective contact pad of the contact pads, wherein a core of the coil includes the MMC filler.
9. The module of claim 1, wherein the IC die includes switching converter components.
10. A switching converter device, comprising: an integrated circuit (IC) die including switching converter components; a leadframe having contact pads; solder elements coupled between the IC die and the contact pads; an insulative layer between the solder elements; a coil having a core and first and second ends, the first end coupled to a first solder element of the solder elements and a respective contact pad of the contact pads, the second end coupled to a second solder element of the solder elements and a respective contact pad of the contact pads; and magnetic-molding compound (MMC) filler in the core and between the IC die and the leadframe.
11. The switching converter device of claim 10, wherein a thickness of the MMC filler between the IC die and the leadframe is limited by a thickness of the insulative layer.
12. The switching converter device of claim 11, wherein the thickness of the insulative layer is based on a target gap between the IC die and the leadframe.
13. The switching converter device of claim 12, wherein the target gap is selected to limit MMC filler particle size between the IC die and the leadframe to less than a threshold particle size.
14. The switching converter device of claim 10, wherein the insulative layer comprises solder resist.
15. A method of manufacturing a module, the method comprising: obtaining an integrated circuit (IC) die; obtaining a leadframe having contact pads; obtaining a coil having a core and first and second ends; adding an insulative layer to partially cover the contact pads; adding solder elements to exposed portions of the contact pads; and adding a magnetic-molding compound (MMC) filler in the core and between the IC die and the leadframe.
16. The method of claim 15, further comprising selecting a thickness of the insulative layer based on a target gap between the IC die and the leadframe.
17. The method of claim 16, further comprising selecting the target gap between the IC die and the leadframe to limit MMC filler particle size to less than a threshold particle size.
18. The method of claim 15, wherein adding the insulative layer comprises: adding solder resist over the contact pads; and etching the solder resist to partially expose the contact pads.
19. The method of claim 15, further comprising adding a polyimide (PI) layer to a side of the IC

die that will face the leadframe.

20. The method of claim 19, further comprising heating the module to reflow the solder elements before adding the MMC filler.
